

Chapter 2: Building a Field-Ready IoT Smart Farming Prototype: Circuit Design, Embedded Firmware, and Multi-Season Deployment Evidence from Smallholder Farms in Pune, Maharashtra

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Abstract

Turning the idea of Internet-of-Things (IoT) precision agriculture into a working device that a smallholder farmer can actually operate receives surprisingly little attention in the literature. Most studies either theorise about benefits or report results from commercial systems far beyond the reach of marginal farmers. This chapter bridges that gap by documenting the full design and field deployment of an original low-cost IoT prototype built around an ESP32 microcontroller integrated with a DHT11 ambient sensor, a DS18B20 soil-temperature probe, a capacitive soil-moisture sensor, a SIM800L GSM module, a 16×2 I2C LCD, a relay-driven irrigation pump, and an active buzzer. The firmware—written in C++ on the Arduino framework—performs real-time sensing, threshold-based automation, SMS alerting, NTP-synchronised CSV logging to on-chip SPIFFS storage, and a Wi-Fi dashboard served from the ESP32 itself. The system was deployed on twelve smallholder farms in Pune district across two crop seasons (Kharif 2024 and Rabi 2024–25). Paired with a survey of 120 farmers, the study records significant gains in crop yield (+27.3%), reduced irrigation water (−34.6%) and net seasonal income (+INR 18,420 per farm), alongside measurable improvements in women’s decision-making autonomy and digital literacy.

Keywords: ESP32; IoT prototype; smart irrigation; SIM800L; precision agriculture; smallholder farming; socio-economic impact

Ethical statement: All household surveys and field measurements were conducted with the informed written consent of participating farmers; no personally identifiable information is disclosed, and the design was approved by the affiliated university’s research ethics committee. Anonymised sensor data and the survey instrument are available from the author on reasonable request.

1. Introduction

Let me be direct about something I have observed repeatedly at the intersection of embedded systems and rural development: the gap between what a sensor-based farming system looks like in a conference paper and what it looks like when a sixty-year-old onion farmer in Pune is expected to operate it is enormous. Many published IoT-agriculture prototypes, once you look past the abstract, were tested in the laboratory—on potted plants, with continuous USB power, stable Wi-Fi, and a graduate student on standby. That is not a farm, and it is not a technology that will change a livelihood.

This chapter does things differently. It documents—completely and honestly—the design, assembly, firmware, and field deployment of an IoT prototype that was built from components available in any electronics market, carried to twelve farms in the Haveli and Khed talukas of Pune district, and left to run through two full crop cycles in Maharashtra’s heat, dust, and monsoon. The data are real field data; the problems were real problems; and the socio-economic outcomes were measured through structured surveys of 120 households, not estimated from simulation. The prototype is built around the ESP32, an affordable dual-core processor with built-in Wi-Fi (Espressif Systems, 2023), and integrates a DHT11, a

DS18B20 soil probe, a capacitive moisture sensor, a SIM800L GSM module, a relay for pump control, a 16×2 I2C LCD, and a buzzer.

2. Related Literature and Positioning

Published work on IoT in agriculture has expanded dramatically since about 2018. A structured review by Friha et al. (2021) surveyed 310 papers and found that most addressed monitoring or automation in isolation, with fewer than 15% reporting both hardware implementation and measurable field outcomes together—an integration deficit this chapter sets out to address. Within the hardware landscape, the ESP32 has become a preferred research platform owing to its dual-core processor, integrated Wi-Fi and Bluetooth, and a retail price consistently below USD 5 (Espressif Systems, 2023). Malavade and Akulwar (2016) demonstrated an early Wi-Fi monitoring node using the ESP8266 in an Indian context, but the ESP32’s richer peripheral set is far better suited to concurrent multi-sensor acquisition.

Component choices were made deliberately. The DS18B20 one-wire probe offers $\pm 0.5^{\circ}\text{C}$ accuracy over $0\text{--}70^{\circ}\text{C}$ with a waterproof form factor (Maxim Integrated, 2019); the DHT11, though less precise than the DHT22, keeps per-node cost low where sensors must be replicated (Aosong Electronics, 2022). A capacitive moisture sensor was preferred over the resistive type because resistive probes corrode within weeks of continuous soil exposure, whereas capacitive units measure dielectric permittivity without passing current through the soil (Decagon Devices, 2016). For alerting, dependence on internet push notifications is a fragility field conditions do not tolerate; the SIM800L delivers SMS wherever a basic 2G signal exists (SIMCom Wireless Solutions, 2015), and as of 2023 some 2G network coverage reached roughly 97% of inhabited Indian villages (Telecom Regulatory Authority of India, 2023).

Field evidence on economic impact remains sparse but is accumulating. Sharma et al. (2020) reported yield improvements of 18–32% from soil-moisture-guided irrigation among vegetable farmers in Andhra Pradesh, and Reddy and Kumar (2025) documented a 24% reduction in irrigation labour among chilli farmers in Telangana. On the social side, digital tools can redistribute decision-making agency within households—particularly to women—but only when training is deliberately gender-inclusive (Moyo & Salawu, 2021). Adoption barriers cluster into economic, technical, and social constraints (Moyo & Salawu, 2021; Paustian & Theuvsen, 2022); this deployment attempts to address each explicitly.

3. Study Area and Participant Selection

The study covered four villages—Koregaon Bhima, Wadgaon Shinde, Nighoje, and Chakan—in the Haveli and Khed talukas of Pune district (approx. 18.52°N , 73.85°E). The region is semi-arid tropical, with mean annual rainfall near 720 mm concentrated in the June–September monsoon, and predominantly Vertisol (black-cotton) soils. Average farm size is 1.2 ha, with about 0.8 ha irrigated. One hundred and twenty farmers were selected by purposive-cum-stratified random sampling: 60 IoT adopters and 60 matched controls, with a minimum 30% women operators among adopters. At baseline the cohorts did not differ significantly on farm size, primary crop, soil type, or income.

Table 1. Socio-Demographic Profile of Study Participants at Baseline

Characteristic	IoT Adopters (n=60)	Control (n=60)
Mean total farm size (ha)	1.18	1.22
Mean irrigated area (ha)	0.79	0.81
Women farmers (%)	33.3%	30.0%
Mean age (years)	42.6	44.1

Characteristic	IoT Adopters (n=60)	Control (n=60)
Mean schooling (years)	9.2	8.7
Smartphone ownership (%)	78.3%	61.7%
Primary Kharif crop: Onion (%)	41.7%	40.0%
Primary Rabi crop: Wheat (%)	48.3%	50.0%

4. Hardware Design: Components, Wiring, and Power

I will walk through the hardware the way I would walk a postgraduate student through it on the bench—with the reasoning behind each choice made explicit, because a circuit description that says what is connected but not why is of little use to anyone replicating it.

Microcontroller. The ESP32 DevKit is powered at 5 V through VIN; its regulator supplies the 3.3 V logic rail. A practical constraint: ADC2 channels are unavailable while Wi-Fi is active, so the moisture sensor is placed on GPIO 33 (ADC1). The dual cores let the web server and sensor loop share time without one starving the other.

Ambient and soil sensing. The DHT11 (GPIO 14, $\pm 2^{\circ}\text{C}$, $\pm 5\%$ RH) needs a 4.7 k Ω pull-up on its data line, the most common student wiring error—without it the library returns NaN. The DS18B20 soil probe (GPIO 4) is buried at 10 cm with a 4.7 k Ω pull-up on DQ; a -127°C reading is the standard open-circuit signature used to spot failed nodes in the field. The 37.0°C high-temperature alert is grounded in plant physiology rather than chosen arbitrarily (Hatfield & Prueger, 2015).

Moisture, GSM, display, relay, buzzer. The capacitive moisture sensor on GPIO 33 outputs an analogue voltage mapped to a percentage (dry ≈ 4095 , wet ≈ 1500 in our calibration); replicators must recalibrate for their own soil. The single most important hardware constraint is the SIM800L's power: it needs a stable 3.7–4.2 V supply able to deliver up to 2 A during transmission bursts, so it is fed from a dedicated 18650 Li-Ion cell—never the ESP32 rail, a mistake that causes resets and corrupted SMS. The 16 $\times$ 2 LCD (I2C, default 0x27) gives an on-device reading without a phone; the relay (GPIO 32) must be wired to COM–NO, not COM–NC; and a 3.3 V active buzzer (GPIO 26) signals threshold breaches.

Table 2. Complete Pin Assignment Reference — ESP32 Smart Agritech Prototype

Component	GPIO	Interface	Power	Key wiring note
DHT11 (air temp/humidity)	14	Digital 1-wire	3.3 V rail	4.7 k Ω pull-up; missing = NaN
DS18B20 (soil temp)	4	OneWire	3.3 V rail	4.7 k Ω pull-up; -127°C = open
Capacitive soil moisture	33 (ADC1)	Analogue	3.3 V rail	ADC1 needed (Wi-Fi disables ADC2)
SIM800L GSM	16 RX / 17 TX	UART2 @ 9600	18650, 3.7–4.2 V	Never power from ESP32 rail
LCD 16 $\times$ 2 (I2C backpack)	21 SDA / 22 SCL	I2C @ 0x27	3.3 V rail	Some modules use 0x3F
Relay (pump control)	32	Digital, active HIGH	5 V adapter	Wire COM–NO only
Active buzzer (3.3 V)	26	Digital, active HIGH	GPIO logic	Must be 3.3 V-rated type
Water pump	via relay	Relay-isolated	12 V / 220 V	No direct ESP32 connection

Table 3. System Power Budget and Supply Architecture

Component	Voltage	Peak current	Supply in this deployment
ESP32 DevKit	5 V (VIN)	500 mA	5 V / 2 A USB adapter
DHT11 / DS18B20	3.3 V	< 5 mA	ESP32 3V3 pin
Soil moisture sensor	3.3 V	< 20 mA	ESP32 3V3 pin
LCD with I2C backpack	3.3 V	< 30 mA	ESP32 3V3 pin
SIM800L GSM	3.7–4.2 V	Up to 2 A (TX)	Dedicated 18650 (2600 mAh)
Relay module	5 V	< 100 mA	5 V USB (shared with VIN)
Active buzzer	3.3 V	< 30 mA	ESP32 GPIO 26
Water pump (12 V DC)	12 V DC	1–3 A	Separate 12 V / 3 A adapter via relay

5. Firmware Architecture and Key Design Decisions

Good field firmware is not the same as code that works in the lab. Field code must run for months, recover from network outages, and report to a farmer standing in the sun who has no time to debug a crash. The firmware is written in C++ on the Espressif Arduino Core (v2.x) and uses standard open-source libraries—LiquidCrystal_I2C (Rickman, 2019), the Adafruit DHT library (Adafruit Industries, 2023), OneWire (Stoffregen, 2019), and DallasTemperature (Burton & Newsome, 2019)—together with the core WiFi, WebServer, SPIFFS, and HardwareSerial modules.

The setup() routine is ordered to fail safe: GPIO outputs are set to safe defaults (pump and buzzer off) before any sensor is read; SPIFFS is mounted but a mount failure does not stop the system; Wi-Fi is attempted for 15 seconds and, if it fails, the node continues in offline mode with sensing, automation, and SMS intact—only the dashboard and NTP timestamps are lost. The loop() interleaves six tasks using non-blocking millis() timing: serving HTTP requests, updating sensors, evaluating thresholds (guarded by an isnan() check), handling incoming SMS queries, alternating the LCD every three seconds, and logging one timestamped CSV row to SPIFFS every five minutes.

Three thresholds are configurable from the dashboard without recompiling: a high soil-temperature alarm (default 37.0°C, with 1.5°C hysteresis), a low-moisture trigger (default 10%) that starts the pump and sends an SMS, and a moisture target (default 70%) that stops it. Boolean flags prevent duplicate SMS within one breach episode. An SMS query feature returns a five-field farm status to authorised numbers; in practice several farmers used it at night from home to confirm the pump had stopped without walking to the field. The dashboard—five live cards, three threshold forms, and CSV download—is stored entirely in PROGMEM and needs no external server.

6. Field Deployment Outcomes

Twelve nodes were installed across the four villages in March–April 2024, each in an IP65 enclosure on a field-edge post with the moisture and DS18B20 probes buried at 10 cm, powered by a 10 W solar panel and a lithium battery pack. Monitoring ran for about thirteen months across Kharif 2024 and Rabi 2024–25, with adopter and control farmers surveyed at baseline, mid-season, and endline.

Harvest data came from plot-level crop-cut surveys. Adopter farms showed consistent, statistically significant yield advantages across all crops and both seasons; for onion, the most important Kharif crop here, adopters averaged 18.7 t/ha against 14.3 t/ha for controls ($t(118) = 7.21$, $p < 0.001$). The composite mean improvement was 27.3%.

Table 4. Crop Yield Comparison — IoT Adopter Farms vs. Control Group

Crop (Season)	Adopters (mean $\pm$ SD)	Control (mean $\pm$ SD)	Difference	p
Onion — Kharif 2024 (t/ha)	18.7 $\pm$ 2.3	14.3 $\pm$ 2.8	+4.4 (+30.8%)	< 0.001
Pearl millet — Kharif (q/ha)	29.4 $\pm$ 3.1	23.6 $\pm$ 3.5	+5.8 (+24.6%)	< 0.001
Groundnut — Kharif (q/ha)	18.1 $\pm$ 2.0	14.8 $\pm$ 2.2	+3.3 (+22.3%)	< 0.001
Wheat — Rabi (q/ha)	38.2 $\pm$ 3.6	30.6 $\pm$ 3.9	+7.6 (+24.8%)	< 0.001
Chickpea — Rabi (q/ha)	14.6 $\pm$ 1.8	11.9 $\pm$ 2.0	+2.7 (+22.7%)	< 0.001
Composite mean yield gain	—	—	+27.3%	< 0.001

Moisture-threshold irrigation replaced calendar-based scheduling. Adopters applied 423 mm in Kharif and 310 mm in Rabi against 647 mm and 473 mm for controls—a mean seasonal reduction of 34.6%, or roughly 38,400 kilolitres saved across the twelve nodes over both seasons (computed on the mean irrigated area of 0.8 ha per holding).

Table 5. Irrigation Water Use and Efficiency — Both Seasons

Parameter	Kharif 2024	Rabi 2024–25	Aggregate / Mean
Irrigation water — adopters (mm)	423	310	366.5
Irrigation water — control (mm)	647	473	560.0
Water saving per season (%)	34.6%	34.5%	34.6%
Irrigation events — adopters	4.2	3.8	4.0
Irrigation events — control	7.1	6.4	6.75
Aggregate water saved, 12 nodes (kL)	21,600	16,800	38,400

Net farm income—gross revenue minus all variable input costs—was the primary economic outcome. Indistinguishable at baseline ($t(118) = 0.42$, $p = 0.67$), the cohorts diverged sharply by endline: adopters reported a mean seasonal net income of INR 94,360 against INR 75,940 for controls, an increment of INR 18,420 ($t(118) = 9.17$, $p < 0.001$). At a per-node hardware cost of INR 3,500 plus about INR 600 per season for connectivity, the simple payback is roughly 3.7 months and the two-season benefit–cost ratio about 4.9:1.

Table 6. Economic Outcome Comparison — Endline Survey (INR per season)

Indicator	Adopters	Control	Difference	p
Mean gross crop revenue	1,34,820	1,12,460	+22,360	< 0.001
Fertiliser and seed cost	18,240	19,620	−1,380	0.04
Irrigation labour cost	4,120	7,850	−3,730	< 0.001
Pesticide cost	6,340	8,250	−1,910	< 0.01
Incremental harvest labour/transport	11,830	870	+10,960	—
Mean net farm income	94,360	75,940	+18,420	< 0.001
Post-harvest loss (%)	8.3%	14.7%	−6.4 pp	< 0.001

Twelve gender-stratified focus groups were held at endline. A notable finding was the effect of SMS querying on women’s agency over irrigation: where scheduling had been a male domain, alerts and remote queries delivered decision-relevant information directly to women’s phones. The Women’s Decision-Making Autonomy Index rose by a mean of 22.4 percentage points among adopters versus 4.1 points among controls (Mann–Whitney $U = 892$, $p < 0.001$). One onion farmer from Wadgaon Shinde put it plainly: “Before this machine I had to ask my husband or the neighbour when to water. Now the phone message tells me directly. I feel I can do this myself.” Digital literacy rose from 43% of adopters

able to navigate the dashboard at baseline to 84% at endline, and household food insecurity fell significantly (Coates et al., 2007).

7. Field Implementation Challenges and Corrective Actions

The most useful service this section can render is a candid account of failures, which published papers rarely document with enough specificity. Two nodes reset during SMS bursts because 26 AWG wiring to the SIM800L cell introduced enough series resistance to sag the voltage; 22 AWG cable and a 470 μ F capacitor across the module's power pins resolved it, and both are now assembly-checklist items. After about four months, three moisture sensors drifted 80–100 ADC counts (a 5–6 point underestimate) and were corrected by reflashing updated map() constants; a browser-based calibration page is planned for v2.0. Extended monsoon overcast cut solar output to 15–25% of rated capacity and took three nodes offline for 6–31 hours—one sited under a neem tree—so future deployments should use ≥ 20 W panels and survey for shading. One node wired to the relay's NC instead of NO ran the pump continuously for 48 hours (about 14,400 L from a self-refilling dug-well); a powered bench-lamp relay test is now mandatory before installation. Finally, three LCDs from a secondary supplier shipped at I2C address 0x3F rather than 0x27, producing blank screens fixed in minutes with a scanner sketch.

8. Scalability Framework and Policy Recommendations

Based on thirteen months of field experience, the following framework is offered for scaling from twelve pilot nodes to the hundreds or thousands needed for regional impact. Each pillar reflects a lesson learned from at least one friction point in this study.

Table 7. Five-Pillar Framework for Scalable Smallholder IoT Deployment

Pillar	What it means in practice	Lesson from this deployment
1. Affordable, repairable hardware	Sub-INR 4,000 nodes from market-available parts	Full node replacement cost INR 3,500
2. Connectivity resilience	SMS over 2G primary; Wi-Fi dashboard secondary	SMS delivered even when Wi-Fi was down
3. Farmer-centred training	Two-day hands-on plus monthly visits; gender-inclusive	78% proficiency by month 3; biggest WDMAI gains where women led
4. Financial access pathways	Self-financing within one crop cycle (3.7-month payback)	41.7% of non-adopters cited upfront cost
5. Ongoing stewardship	Quarterly calibration; OTA updates; village technicians	OTA would have resolved two on-site incidents remotely

Five policy recommendations follow. First, integrate IoT hardware procurement into PM-KISAN by allowing farmers to redeem part of the direct benefit transfer against approved sensor kits, addressing the cost barrier without new outlay. Second, embed soil-moisture sensing as a standard component of every PMKSY micro-irrigation installation. Third, create a dedicated IoT agri-extension cadre within the KVK network—one trained officer per cluster of fifty villages. Fourth, mandate gender-inclusive deployment standards, with quantified targets for women's operational autonomy, in publicly funded agri-tech. Fifth, establish a state-managed, consent-governed open agricultural data repository federated from node data streams to support early-warning, insurance, and climate-resilience planning.

9. Discussion

These results confirm what the precision-agriculture literature has argued theoretically: sensor-guided irrigation and temperature monitoring can deliver substantial yield and income gains even on very small farms, at hardware costs recoverable within a single crop cycle. The composite +27.3% yield gain sits at the upper end of Sharma et al. (2020) and aligns with the labour reductions of Reddy and Kumar (2025). What this study adds is an honest account of what happens between the circuit diagram and the harvest data—five hardware failure modes, a solar vulnerability, a relay error, calibration drift, and an address mismatch—none of which appear in the results tables but all of which shaped them.

The social findings deserve emphasis. The gain in women's autonomy was not an automatic consequence of installing a sensor; it followed from training that included women as primary operators, an SMS format accessible to low-literacy users, and an on-device LCD that removed the need to open an app. Technology design is not socially neutral. The main limitation is geographic concentration: thirteen months across twelve nodes in four villages is strong proof-of-concept, not a representative quantification of average impact, and warrants larger multi-district trials.

10. Conclusion

This chapter has presented the complete design, firmware, and field deployment of an original IoT smart-farming prototype built for under INR 3,500 per node and operated by smallholder farmers across two crop seasons in Pune district. The system was not theorised in a laboratory; it was designed, broken, fixed, reinstalled, recalibrated, and run to harvest. The headline numbers—a 27.3% yield improvement, 34.6% less irrigation water, an INR 18,420 mean seasonal income gain, and a 3.7-month payback—are meaningful, but the more durable contribution is the demonstration that IoT smart farming need not be the preserve of large commercial operations. With careful hardware selection, firmware that prioritises offline resilience, and training that treats farmers as capable operators, a field-hardened precision-agriculture system is achievable within the resource constraints of smallholder farming in Maharashtra. The prototype is a beginning: it still needs over-the-air updates, in-browser recalibration, and testing across more soils and cropping systems—but it worked, it survived thirteen months of monsoon and heat, and the farmers who used it earned meaningfully more than those who did not.

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